

## Memory Connections to CPU

- **RAM and ROM chips** are connected to a CPU through the data and address buses.
- **The low-order lines** in the address bus select the byte within the chips and other lines in the address bus select a particular chip through its chip select inputs.
- **The connection of memory chips** to the CPU is shown in Fig. 4. This configuration gives a memory capacity of 512 bytes of RAM and 512 bytes of ROM. It implements the memory map of Table 1. Each RAM receives the seven low-order bits of the address bus to select one of 128 possible bytes.
- **The particular RAM chip** selected is determined from lines 8 and 9 in the address bus. This is done through a 2 x 4 decoder whose outputs go to the CS1 inputs in each RAM chip.
- **Thus, when address** lines 8 and 9 are equal to 00, the first RAM chip is selected. When 01, the second RAM chip is selected, and so on. The RD and WR outputs from the microprocessor are applied to the inputs of each RAM chip.

- **The selection between** RAM and ROM is achieved through bus line 10. The RAMs are selected when the bit in this line is 0, and the ROM when the bit is 1.
- **The other chip select** input in the ROM is connected to the RD control line for the ROM chip to be enabled only during a read operation. Address bus lines 1 to 9 are applied to the input address of ROM without going through the decoder.

**TABLE 1** Memory Address Map for Microprocomputer

[illegible]

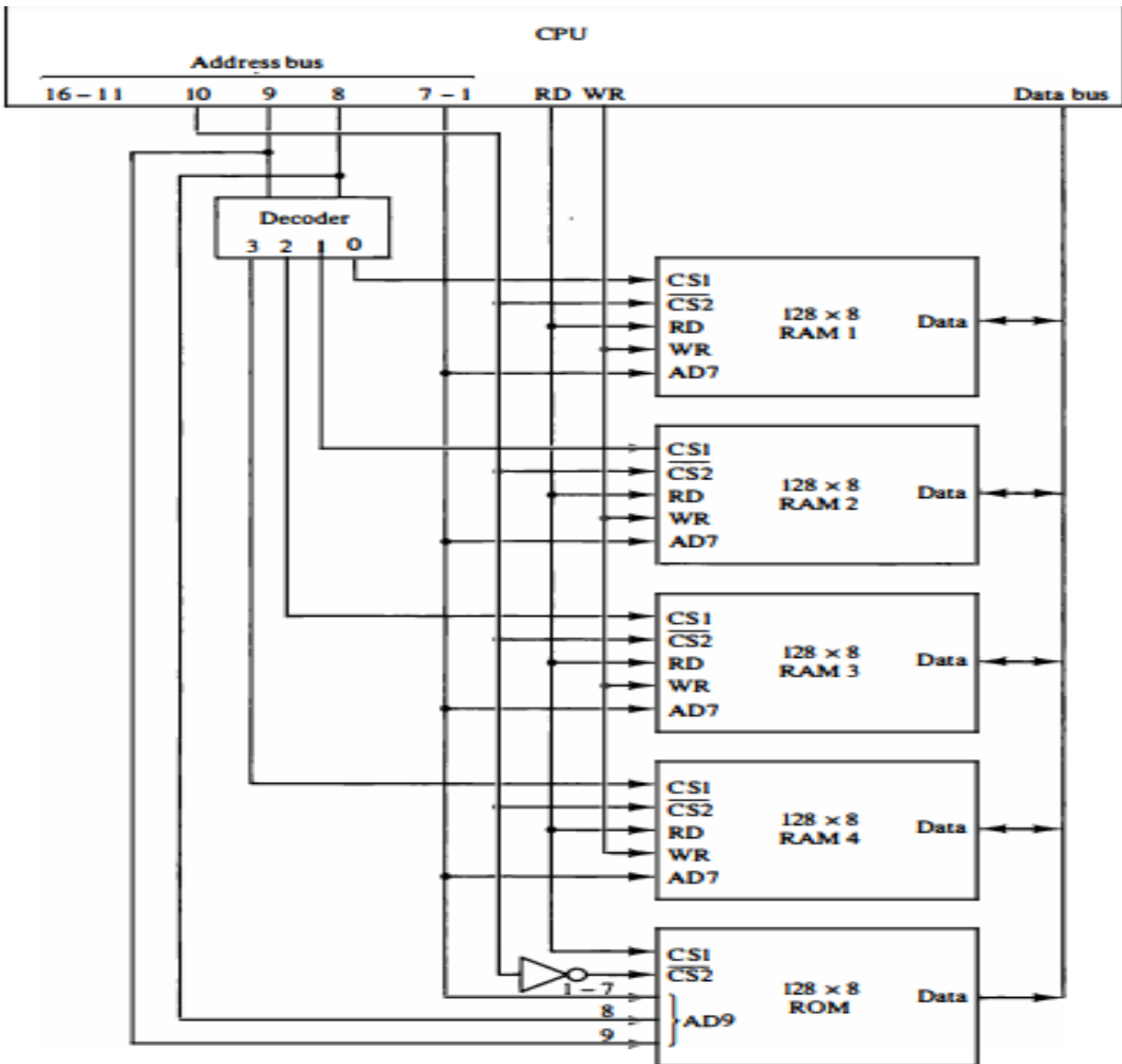


Figure 4 Memory connection to the CPU.